AVAC4QGIS User Interface Guide

Interface reference for AVAC4QGIS 0.5.12

LHE–EPFL, Switzerland

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This guide describes the controls that are visible in the packaged QGIS plugin. It was checked against the current implementation in `avac_qgis/gui/dock.py` and the preparation, coupling, and result modules used by those controls. Hidden compatibility fields and developer-only code paths are not presented as user features.

Contents

1 General description of the interface

AVAC4QGIS is a dockable QGIS panel for preparing, running, and inspecting depth-averaged avalanche simulations. The standard workflow has three tabs: **AVAC Parameters**, **AVAC Run**, and **Results**. Inputs, scientific parameters, execution, visualization, profiles, gauges, and PNG exports remain inside QGIS, while packaged native solvers run in isolated case directories.

The optional **Enable Lake-Wave Extension** check box inserts **WAVE Parameters** and **WAVE Run**. WAVE consumes a completed AVAC run as a read-only source and uses the same final Results tab. Enabling WAVE does not change an existing AVAC run.

The parameter and result pages use accordion controls. Click an accordion header to display its contents. Each major operation reports its state through status text and a progress bar. Long execution logs remain hidden until their corresponding log button is enabled.

2 General simulation workflow

2.1 AVAC-only simulation

1. Choose a writable **AVAC Working Directory** above the tabs.
2. In **AVAC Parameters**, select the terrain DEM and release polygons, then load or set the release, rheology, and numerical values.
3. In **AVAC Run**, use **Check Environment** and **Validate Inputs**. Use the previews if the initial state or altitude-dependent rheology must be inspected.
4. Select **Prepare AVAC Run**. Preparation creates a new, self-contained run below `runs/`; it does not execute the solver.
5. Select **Run**. A completed run becomes available in **Results**.
6. In **Results**, select the run and load summary or temporal products. Profiles, gauges, and exports load a missing temporal product automatically before continuing.

2.2 AVAC followed by WAVE

1. Complete AVAC first and enable the Lake-Wave Extension.
2. In **WAVE Parameters**, choose the completed AVAC run, terrain DEM, lake water-body polygon, water-surface elevation, WAVE cell size, and model settings. The lake polygon may instead be created from the DEM, water elevation, and a map click before AVAC has been run.
3. Use **Preview Water Level**. The exact solver-grid lake state is cached and reused by preparation while its inputs remain unchanged.
4. In **WAVE Run**, validate and prepare the WAVE scenario. WAVE inherits the completed AVAC calculation bounds, duration, output count, and temporal origin. Preparation creates the internal shoreline source without modifying AVAC output.
5. Run WAVE and return to the shared Results tab. Select both the completed AVAC run and its completed WAVE scenario when comparing snow and water products.

> WAVE and AVAC both solve on rectangular GeoClaw grids, but the snow-to-water transfer is not imposed at the outer rectangle. It is applied internally on the grid-aligned shoreline of the initially wet lake. Only AVAC flux directed into the wet lake contributes to the WAVE source.

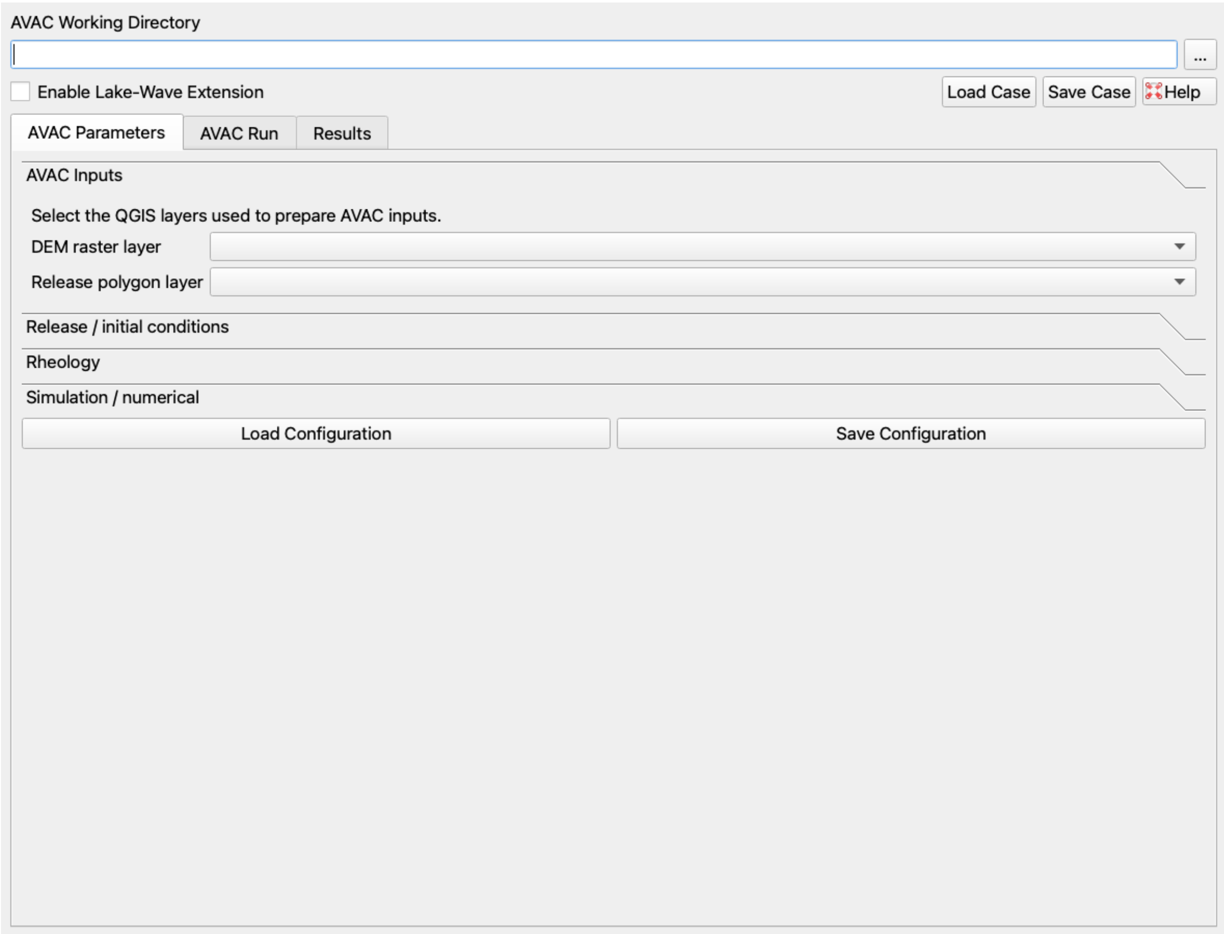

Figure 1: The complete dock with the persistent controls above the tabs and the normal AVAC workflow.

3 Function of each interface element

This section follows the current visual order of the interface. It describes what each visible element reads, changes, or creates.

3.1 Controls above the tabs

- **AVAC Working Directory** is the required writable workspace. Type a path or use the browse button. It is user data storage, not the plugin installation or the managed runtime. QGIS remembers the path.
- **Enable Lake-Wave Extension** shows or hides WAVE Parameters and WAVE Run and adds WAVE variables and Lake Volume to Results. It does not alter AVAC files.
- **Load Case** restores a complete AVAC4QGIS case YAML: workspace, persistent input-layer sources, visible AVAC values, WAVE enabled state, and WAVE setup when present.
- **Save Case** writes the corresponding complete case YAML. It stores references to persistent layer sources; temporary memory layers must be saved to disk before the case can be portable.
- **Help** opens this PDF. The Plugins > AVAC4QGIS menu also contains Help. Its online-documentation and website entries are present but disabled until those destinations exist.

3.2 AVAC Parameters tab

This tab contains four accordion pages and the AVAC-only configuration buttons. Changing a visible input or parameter invalidates any prepared run; prepare again before executing it.

3.2.1 AVAC Inputs

- **DEM raster layer** selects the fixed basal topography. The first raster band must have a valid CRS, finite usable cells, and square pixels. The selected DEM extent defines the rectangular AVAC calculation domain. The terrain resolution must equal or evenly subdivide the chosen computational cell size.
- **Release polygon layer** selects polygon or multipolygon features defining the initially mobile release area. Geometries are transformed to the DEM CRS during preparation when needed. At least one DEM grid point must fall inside the combined polygons.

Preparation copies the selected sources into the run's `inputs/` tree for traceability. The optional refinement-DEM backend remains in the source for compatibility but has no visible selector and is not part of the normal workflow.

3.2.2 Release / initial conditions

The DEM remains the fixed basal surface. These controls initialize only the mobile release depth. There is no additional stationary winter snowpack: outside the release polygons, initial mobile depth is zero and the initial snow-surface product equals the DEM.

- **Release depth** d_0 (m) is the base mobile thickness. With both corrections disabled, every release-grid point receives d_0.
- **Critical slope** θ_{cr} (degrees) and **Slope correction** ν define the optional local-slope factor. The implementation evaluates it only where DEM slope exceeds 25° and the denominator is numerically safe.

- **Reference elevation** z_{ref} (m) and **Hypsometric gradient** define the optional elevation change $(z - z_{ref})\, g_h/100$. A positive gradient increases depth above z_{ref} and decreases it below z_{ref}.
- **Apply slope correction** and **Apply elevation correction** select which corrections are applied. If both are active, the elevation-corrected depth is multiplied by the slope factor.
- **Return period [years]** is positive integer scenario metadata. It labels the case and results but does not change AVAC equations or the release-depth calculation.

The initial-depth preview should be inspected when corrections are active. The code replaces nonfinite corrected values with zero, but does not silently substitute a different scientific correction model.

3.2.3 Rheology

- **Model** selects **Coulomb**, **Voellmy**, or **cohesive_Voellmy**. Coulomb uses basal dry-friction resistance. Voellmy adds speed-dependent resistance. Cohesive Voellmy adds cohesive strength and its static-yield contribution.
- **Density** ρ (kg/m^3) is bulk avalanche density. It is used by the cohesive term and by the displayed kinetic-pressure result $p_k = \rho v^2/2$, reported in kPa.
- μ is the dimensionless Coulomb-friction coefficient. Increasing it strengthens the speed-independent basal resistance.
- ξ (m/s^2) controls the Voellmy speed-dependent term. In the implemented term, larger ξ means weaker speed-dependent resistance. It is disabled for the Coulomb-only model.
- **Cohesion** C (Pa) adds material strength in the cohesive Voellmy model and is disabled for the other two models.
- **Altitude zones** replaces uniform μ, ξ, and C with bed-elevation zones. Enter one comma-separated row per zone as `mu, xi, cohesion, lower elevation`. The first row must leave the fourth field blank. Later lower elevations must be strictly increasing; a cell exactly at a break belongs to the higher zone.
- **Visualize Rheology Zones** writes `qgis_previews/avac_rheology_zones.tif` and loads it as a categorical raster. Its colors and legend show exactly which zone each valid DEM cell will use. It does not prepare or alter a run.

The AVAC source update integrates the local Coulomb/cohesive and Voellmy resistance with an explicit zero-speed arrested state. This is why rheology is material behavior, not merely result styling.

3.2.4 Simulation / numerical

- **Simulation duration** is the final elapsed solver time in seconds.
- **AMR refinement levels** is the maximum number of adaptive mesh levels. Additional levels can resolve flagged flow regions more finely and increase cost. Level 1 is the unrefined base calculation.
- **Output interval** is the requested elapsed time between outputs. Its default is exactly 1 s. When duration or interval is edited, the plugin calculates a whole number of intervals and includes both $t = 0$ and final time in the temporal raster. A loaded configuration retains its explicit output count until one of these timing controls is changed.
- **CFL target** is the normal adaptive time-step target and **CFL maximum** is the rejection limit. Target must not exceed maximum.

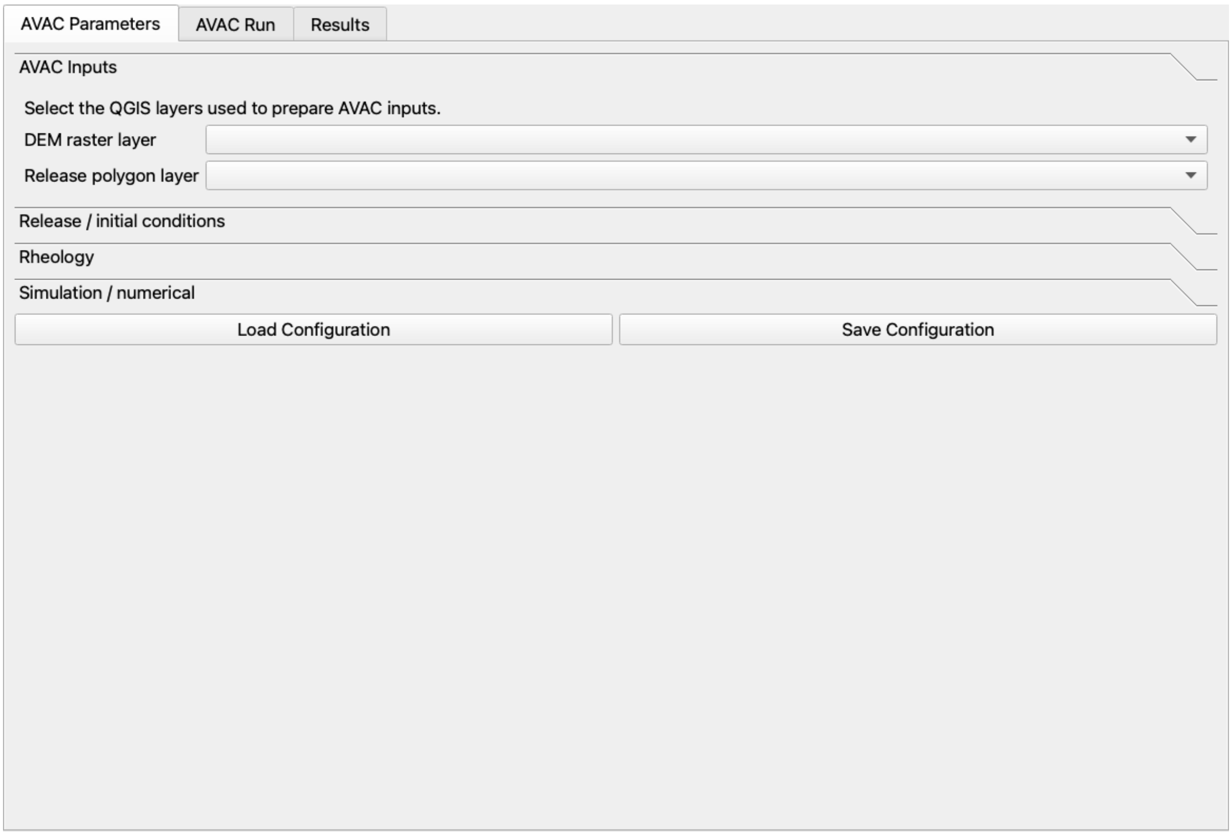

Figure 2: AVAC Parameters with the AVAC Inputs accordion open.

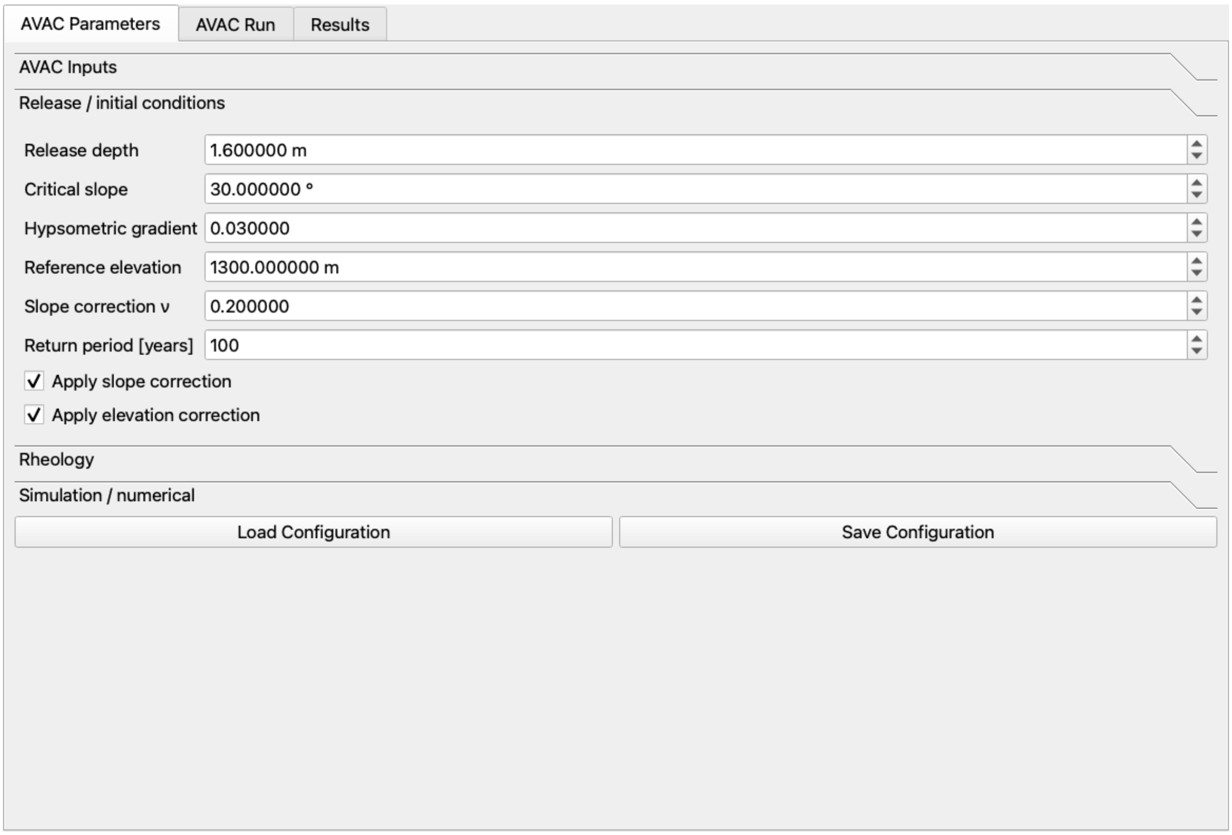

Figure 3: AVAC Parameters with Release / initial conditions open.

- **Computational cell size** (m) is base solver spacing. It must equal or be a whole-number multiple of the DEM pixel size, and both domain spans must contain a whole number of solver cells.
- **Limiter** selects none, minmod, superbee, MC, or van Leer for second-order wave limiting. The AVAC default is superbee.

There is no boundary selector. Every normally prepared AVAC case uses extrapolating outer boundaries, so material may leave the rectangular domain. The prepared YAML also standardizes binary 32-bit output, zero solver verbosity, and Depth as the default temporal variable.

3.2.5 Load Configuration and Save Configuration

Load Configuration reads a complete AVAC YAML schema, restores the visible values, and makes that file the template for preparation. Unknown advanced keys remain in the complete template rather than being discarded. **Save Configuration** applies the current visible values and fixed run choices to that complete schema and writes a new YAML file after validation. These buttons affect AVAC settings only; Load Case and Save Case above the tabs cover the complete plugin setup.

3.3 AVAC Run tab

- **Prepared Simulation** summarizes the isolated run, return period, duration, output interval, cell size, and rheology after preparation.
- **Active run** is read-only and shows the directory that Run will execute.
- **CPU cores** sets `OMP_NUM_THREADS` from one to the maximum logical cores detected on the computer. The default is the maximum.
- **Check Environment** checks the writable workspace, complete built-in or user-selected configuration template, and managed AVAC runtime. External Make, gfortran, Python, and CLAW variables are not required for packaged execution.
- **Validate Inputs** checks DEM, release geometry, visible parameters, CRS, and grid compatibility without creating a run.
- **Prepare AVAC Run** creates a timestamped directory under `runs/`, materializes source inputs, computes the terrain and initial condition, writes the run configuration, and stages runtime data. Its progress bar reports this background task.
- **Preview Initial Depth** loads the exact mobile h_0 field used by preparation. **Preview Initial Snow Surface** loads $z + h_0$. Both reusable rasters are written below `qgis_previews/` in the Working Directory.
- **Run** starts the prepared native solver. **Stop** requests termination of that AVAC process. A stopped run remains on disk but is not marked as completed.
- The execution status and progress bar report readiness and frame progress. **Show execution log** reveals environment, preparation, solver, and error messages.

3.4 WAVE Parameters tab

This tab exists only while the Lake-Wave Extension is enabled. It has three accordion pages. WAVE is a separate water solver and does not write into the selected AVAC run.

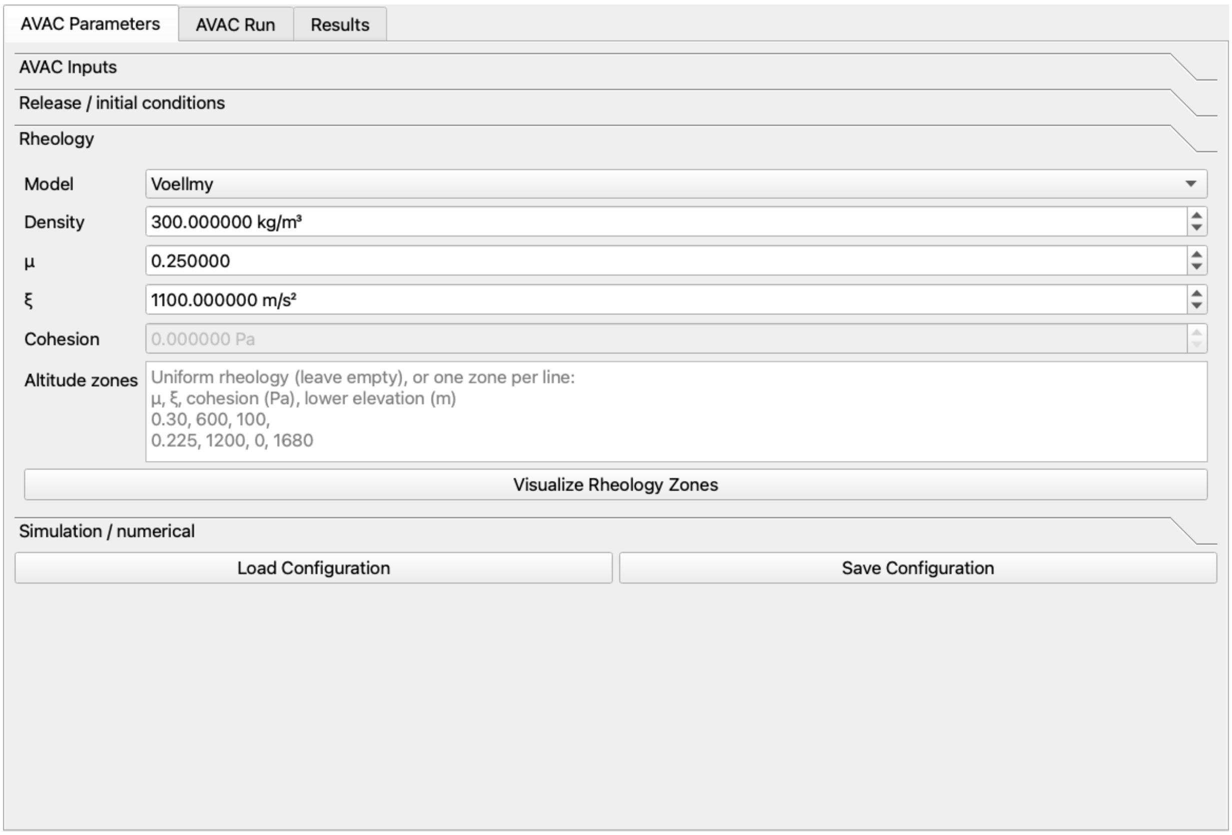

Figure 4: AVAC Parameters with the Rheology accordion open.

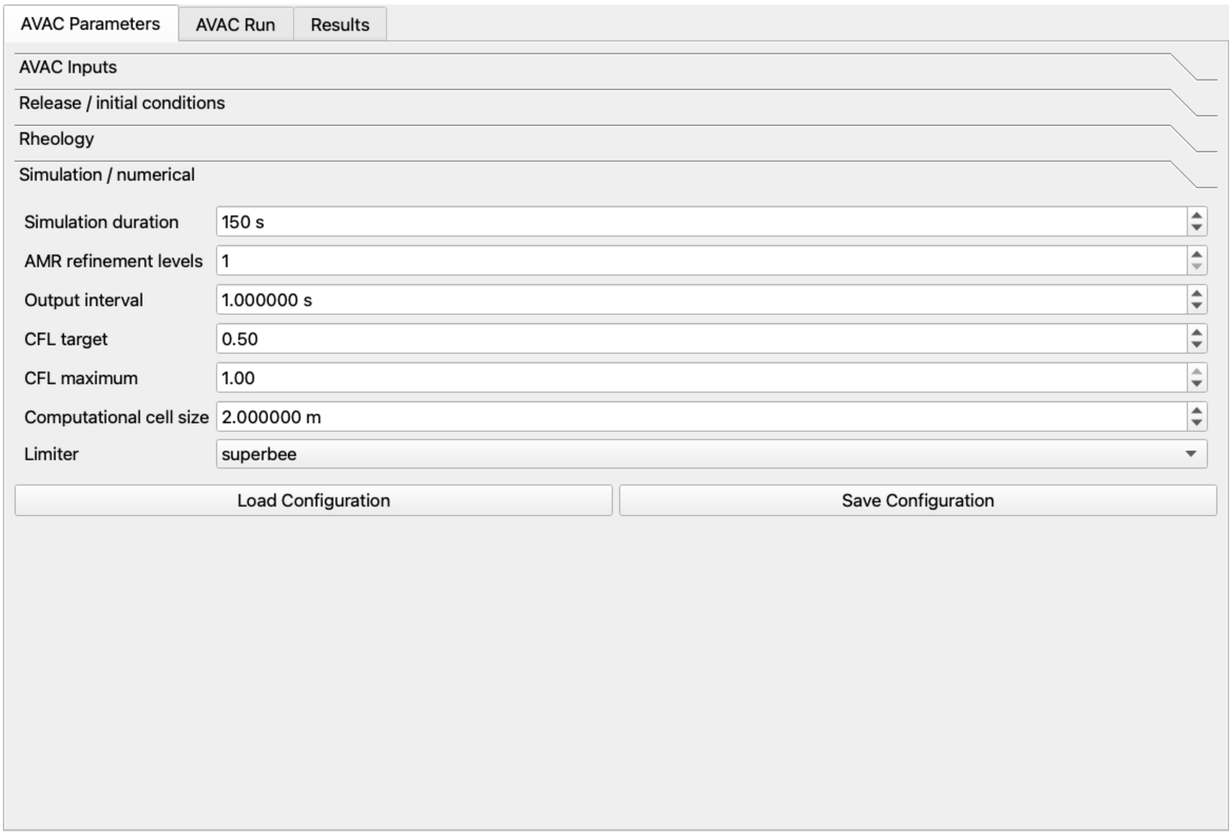

Figure 5: AVAC Parameters with Simulation / numerical open.

3.4.1 AVAC Source

Completed AVAC run lists completed runs in the Working Directory. **Refresh Completed AVAC Runs** refreshes the list. The selected run provides AVAC frames, CRS, rectangular calculation bounds, duration, output count, and the shared temporal origin.

WAVE has no independent calculation-domain controls. Its rectangular solver domain is exactly the selected completed AVAC domain. WAVE may use a different, coarser cell size, but its terrain must cover these bounds and share the AVAC CRS.

3.4.2 Terrain and Lake Inputs

- **Terrain/bathymetry DEM** is the WAVE bed source. It must use square pixels, have a valid CRS, and cover the complete AVAC domain. It is commonly the same source DEM used by AVAC, but it may include bathymetric information not present in the AVAC topography.
- **Lake water-body polygon** defines the actual reservoir or lake footprint. It determines the initially wet mask and the internal shoreline used for AVAC-to-WAVE transfer. It is not the rectangular solver domain.
- **Water level** (m) is absolute initial free-surface elevation in the DEM's vertical datum, not water depth. A cell inside the lake is wet only when its prepared bed is below this elevation. A numeric value of zero is therefore an elevation of 0 m in that datum; it does not mean "empty lake."
- **Create Lake Polygon from Map Point** arms one QGIS map click. From the clicked DEM cell it finds the four-connected cells at or below Water level, expands its DEM read window until the contour closes, and writes a persistent GeoPackage under `derived_inputs/`. It requires the DEM, water level, map point, and writable Working Directory, but no completed AVAC run. A right click cancels selection.
- **Preview Water Level** creates and loads the exact WAVE-grid initial depth. It uses the lake polygon and selected WAVE cell size, stabilizes the grid-aligned shoreline, stores the GeoTIFF under `qgis_previews/`, and caches the prepared lake state.
- **Wave grid cell size** (m) must equal the terrain resolution or be a larger whole-number multiple. Coarsening reduces runtime and memory; unsupported finer or fractional resampling is rejected.

The lake polygon is rasterized consistently with the WAVE mesh. When the DEM is finer, a solver cell becomes lake only when at least half of its fine-grid coverage is inside the polygon. A narrow shoreline guard is adjusted to the waterline plus dry tolerance to prevent artificial waves at $t = 0$. Exterior cells are not permanently dry: resolved runup may wet them later.

3.4.3 WAVE Model Settings

- **Snow-to-water damping** multiplies the internally coupled AVAC source rates. The permitted interface range is 0 to 0.4; smaller values inject less water volume and momentum.
- **Land Strickler coefficient** and **Water Strickler coefficient** set roughness on the two bed-elevation sides of the water-level break. WAVE passes their reciprocals as Manning coefficients, so larger Strickler means weaker Manning resistance.
- **Friction depth limit** is the maximum water depth on which the Manning source term is applied.
- **Dry-depth limit** is the WAVE wet/dry threshold and the small elevation offset used by shoreline stabilization.

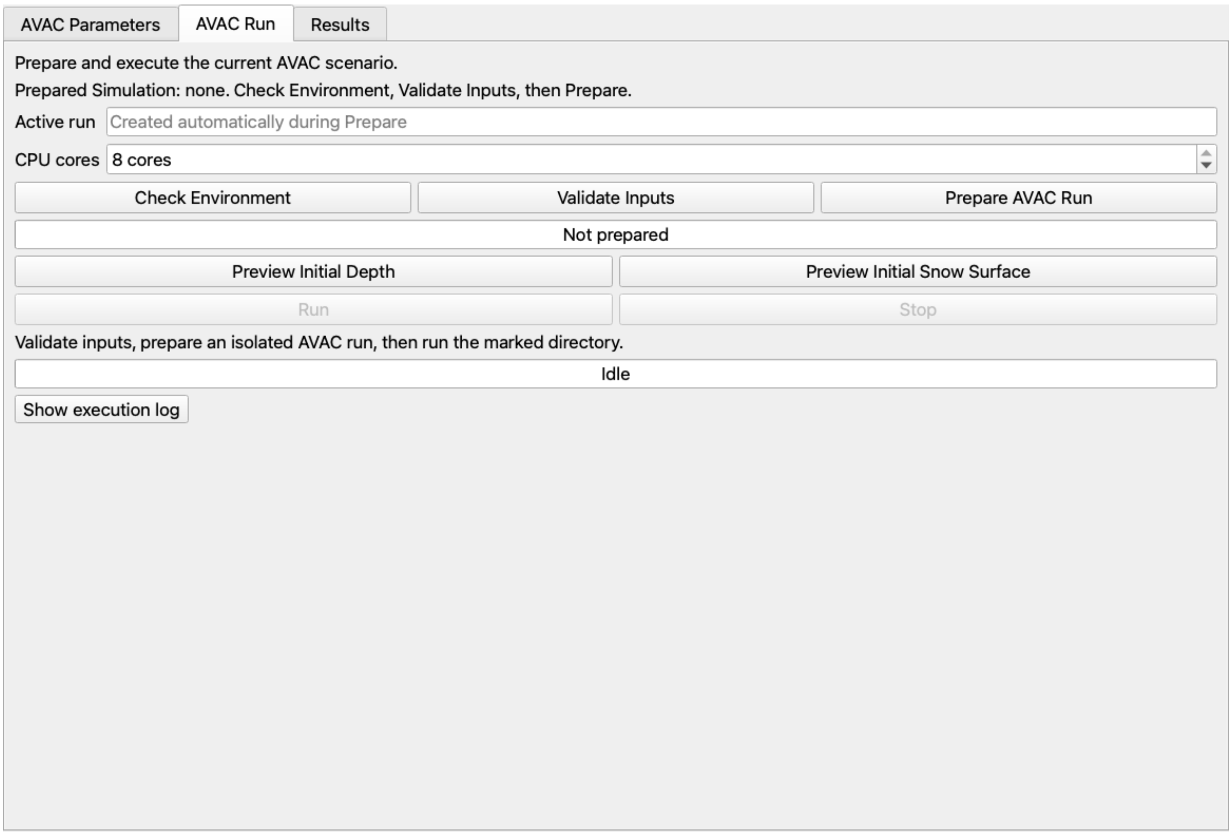

Figure 6: The current AVAC Run tab.

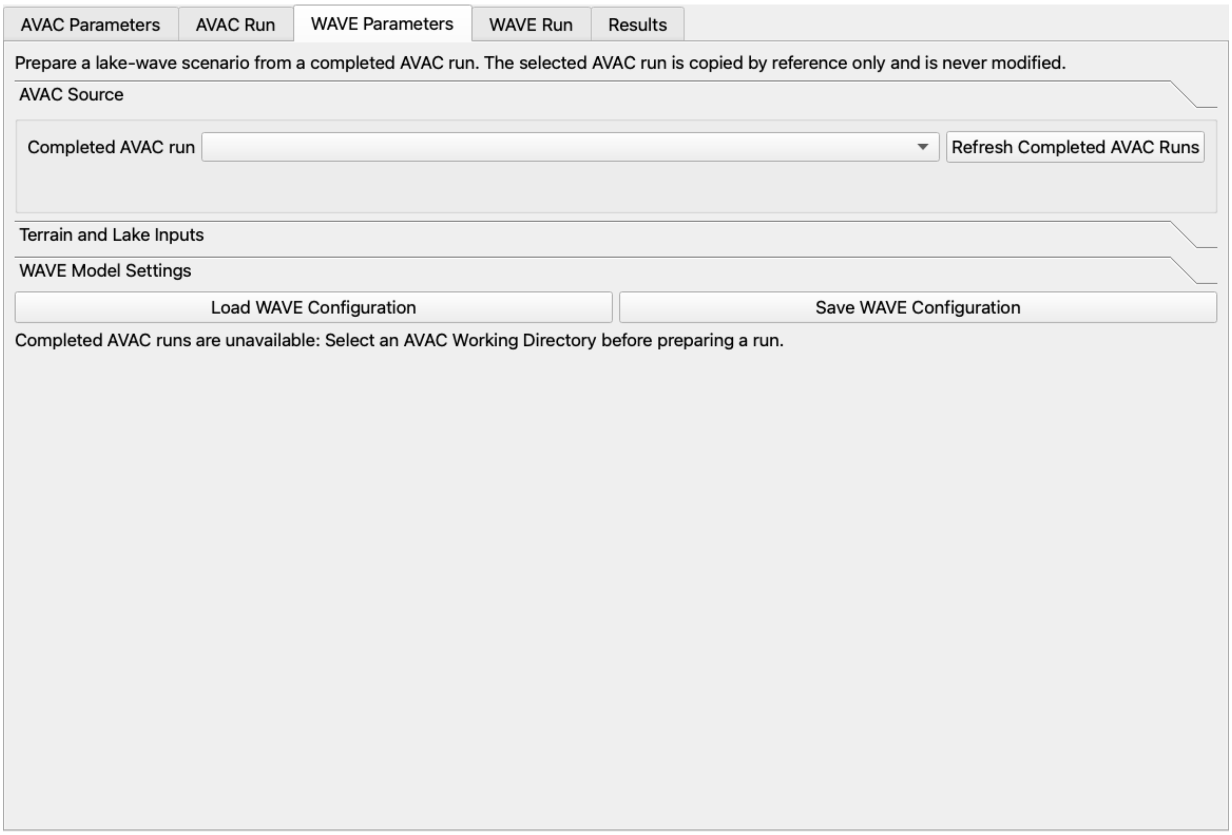

Figure 7: WAVE Parameters with the AVAC Source accordion open.

- **Wave refinement tolerance** controls GeoClaw wave-based AMR flagging. WAVE currently prepares one AMR level in the normal plugin workflow; the tolerance remains part of its configuration.
- **CFL target** and **CFL maximum** control adaptive time stepping. Target must not exceed maximum.
- **Limiter** selects none, minmod, superbee, MC, or van Leer. The WAVE default is van Leer.

Load WAVE Configuration and **Save WAVE Configuration** restore or write only this WAVE setup: selected AVAC source reference, terrain, lake polygon, water level, cell size, and model values. They do not change AVAC Parameters.

3.4.4 Internal AVAC-to-WAVE source

During WAVE preparation, the plugin constructs north/south/east/west faces around the initially wet lake on the WAVE grid. At each AVAC output time it samples the three conserved AVAC components h, hu, and hv at each face. The inward normal flux is

$$q_n = hu\, n_x + hv\, n_y.$$

Only finite samples with positive depth and $q_n > 0$ are active. Multiplication by face length and Snow-to-water damping produces an integrated water-volume rate Q. WAVE receives all three conservative source rates

$$(Q,\ Qu,\ Qv),$$

not mass alone. The WAVE solver divides those integrated rates by the area of the base-grid or AMR cell that receives the point source. This prevents mesh refinement from changing the total injection.

Preparation integrates the source history and records injected water volume, both depth-momentum components, and both physical water-momentum components in `CL/summary_config.yaml` and in the WAVE execution log. If no inward-moving AVAC sample reaches the wet shoreline, preparation reports that the lake was not reached and leaves Run unavailable.

3.5 WAVE Run tab

- **Prepared Simulation** identifies the WAVE scenario and its AVAC source after preparation.
- **Check Environment** verifies the writable workspace, managed WAVE runtime, solver, dependencies, and selected core count.
- **Validate Inputs** checks the completed AVAC source, terrain, lake polygon, inherited AVAC domain, cell-size alignment, CRS, and current water-level preview state without creating a scenario.
- **Prepare Wave Run** creates a timestamped directory below `wave_runs/`, reuses or creates the water-level preview, writes terrain and wet/dry inputs, and creates the internal shoreline source. WAVE duration and output count are inherited from AVAC.
- **Run** starts the prepared WAVE solver. **Stop** first requests a clean termination and force-stops the process after a grace period if needed.
- **CPU cores** sets WAVE's OMP_NUM_THREADS and defaults to the detected maximum.
- Status text and both progress bars report preparation and output-frame progress. **Show Wave execution log** reveals the preparation, coupling ledger, runtime, and solver messages.

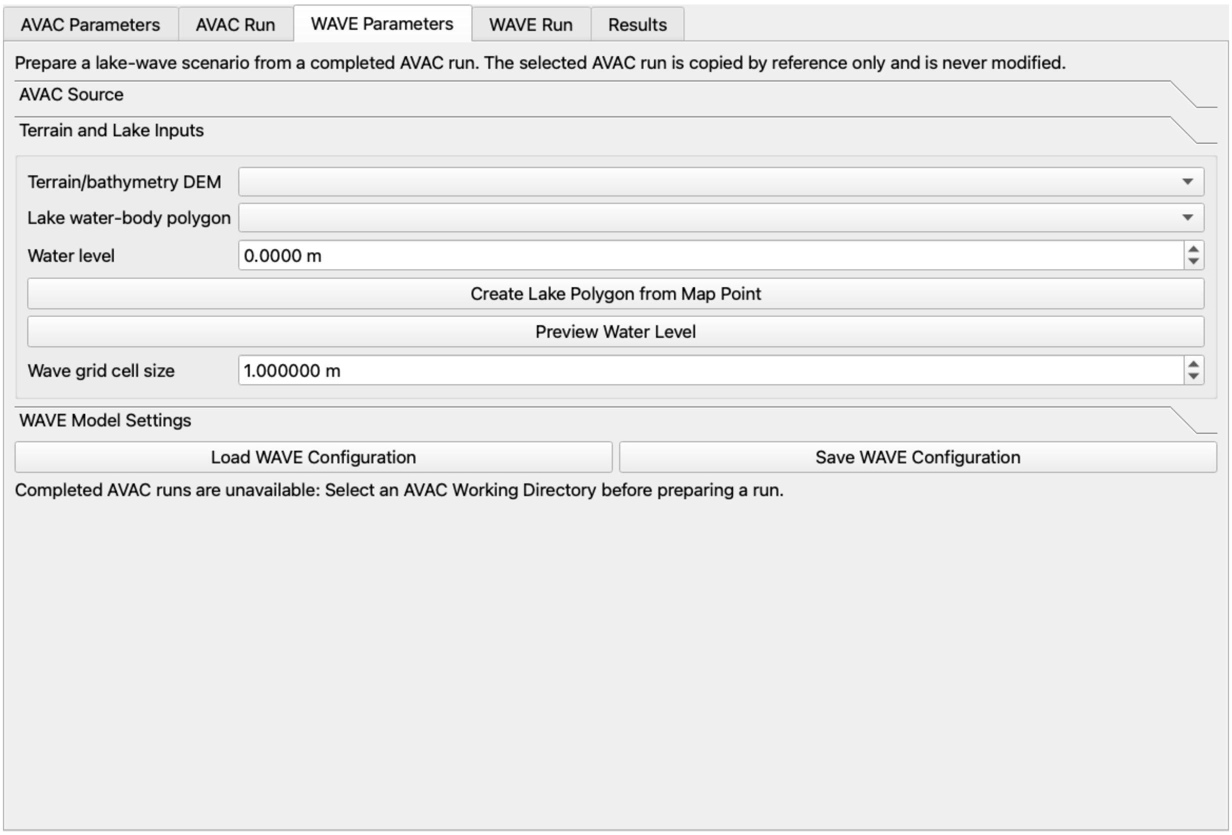

Figure 8: WAVE Parameters with Terrain and Lake Inputs open. The point-derived lake polygon does not require a prior AVAC run.

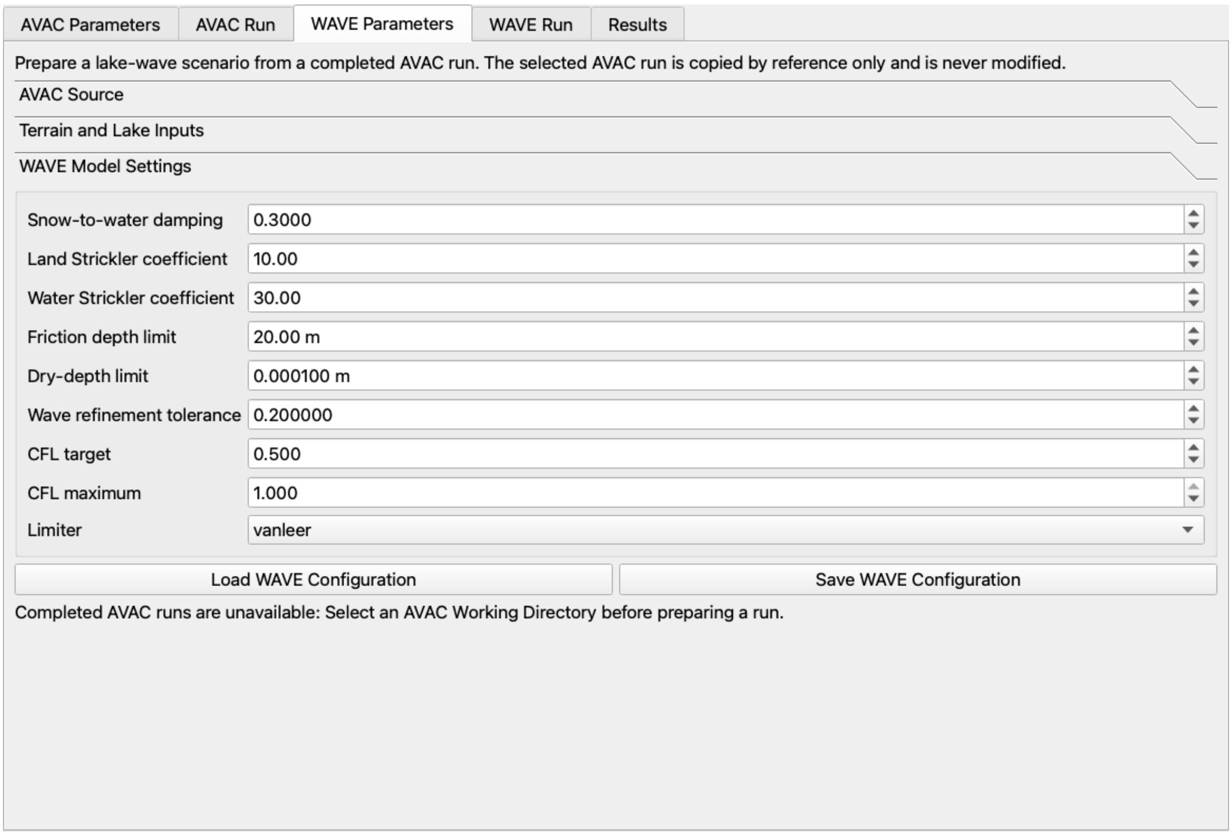

Figure 9: WAVE Parameters with WAVE Model Settings open.

3.6 Results tab

Results is always the last tab. It is shared by both solvers. Controls select a product whose label begins with AVAC or WAVE and dispatch the work to the corresponding completed run. The progress bar remains at the bottom of the tab.

3.6.1 Results Run

- **Completed workspace run** and **Refresh Runs** select a completed AVAC run from the Working Directory.
- **Open Run Directory** selects an external completed AVAC run directory.
- When WAVE is enabled, **Completed WAVE scenario**, its Refresh Runs button, and **Open Wave Scenario Directory** select a completed WAVE scenario. AVAC and WAVE summaries remain visible together.

3.6.2 Summary Maps

AVAC summary choices are Maximum Depth, Maximum Velocity, Maximum Pressure, Arrival Time, Time of Maximum Depth, and Time of Maximum Velocity. Maximum Pressure is the kinetic-pressure display $\rho v^2/2$ in kPa. Time products contain elapsed simulation seconds.

When WAVE is enabled, the selector also contains Maximum Surface Rise and Maximum Surface Drawdown. These are the largest positive rise and largest positive magnitude of drawdown relative to the initial WAVE free surface. **Load Map** creates or reuses the selected GeoTIFF and adds it to QGIS.

3.6.3 Time Series Maps and PNG export

- **Variable** selects AVAC Depth, Velocity, Pressure, or Snow Surface Elevation. With WAVE enabled it also offers WAVE Snow Depth Outside Lake, WAVE Water Depth, Water Elevation, and Surface Displacement.
- **WAVE Snow Depth Outside Lake** is a derived copy of AVAC depth with AVAC cells intersecting the prepared lake footprint set to zero in every band. The original AVAC product is unchanged. Band count and elapsed times are preserved.
- **WAVE Water Elevation** is $B + h$ only on wet cells; dry cells are NoData. **WAVE Surface Displacement** is $(B + h)$ minus the initial surface. Its color ramp is diverging and exactly zero is transparent.
- **Load Time Series** creates or reuses a multiband GeoTIFF and loads it into QGIS. AVAC and WAVE layers inherit one shared simulation origin, so equal elapsed times align.
- The **Frame Player** has a seek slider, Play, Pause, Restart, and frames-per-second control. It directly switches raster bands and does not require tuning QGIS Temporal Controller dates.
- **Include** selects a simulation-time title, legends, and scale bar for PNG output. A legend uses the active layer's QGIS color ramp. If multiple visible temporal layers are exported together, each gets its own legend.
- **Extent** chooses Current map canvas or Selected result extent for rendering only. **Image width** accepts 640–8000 pixels.
- **Export Current Map PNG** renders the current map and frame. **Export Time Series Frames** renders the selected series, and **Export every Nth frame** controls sampling. **Cancel Export** stops the loop; completed PNG files remain.

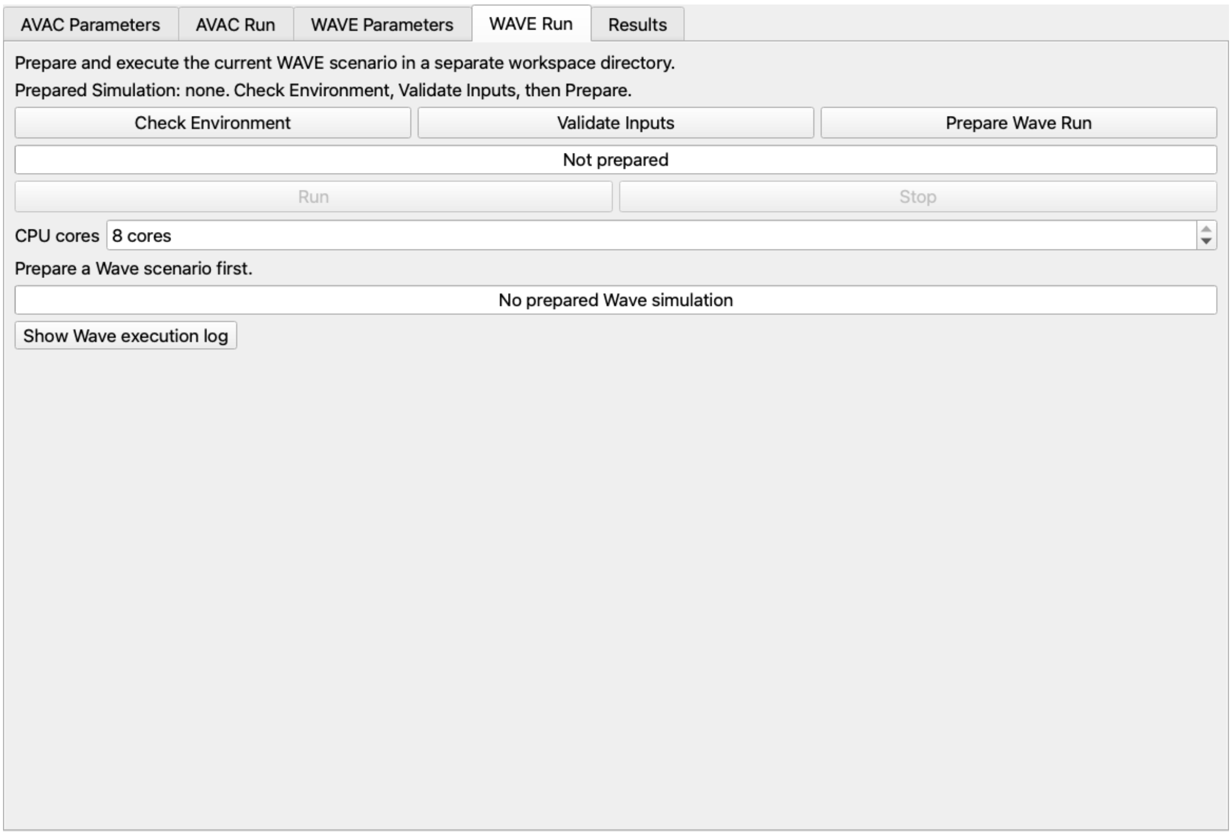

Figure 10: The current WAVE Run tab.

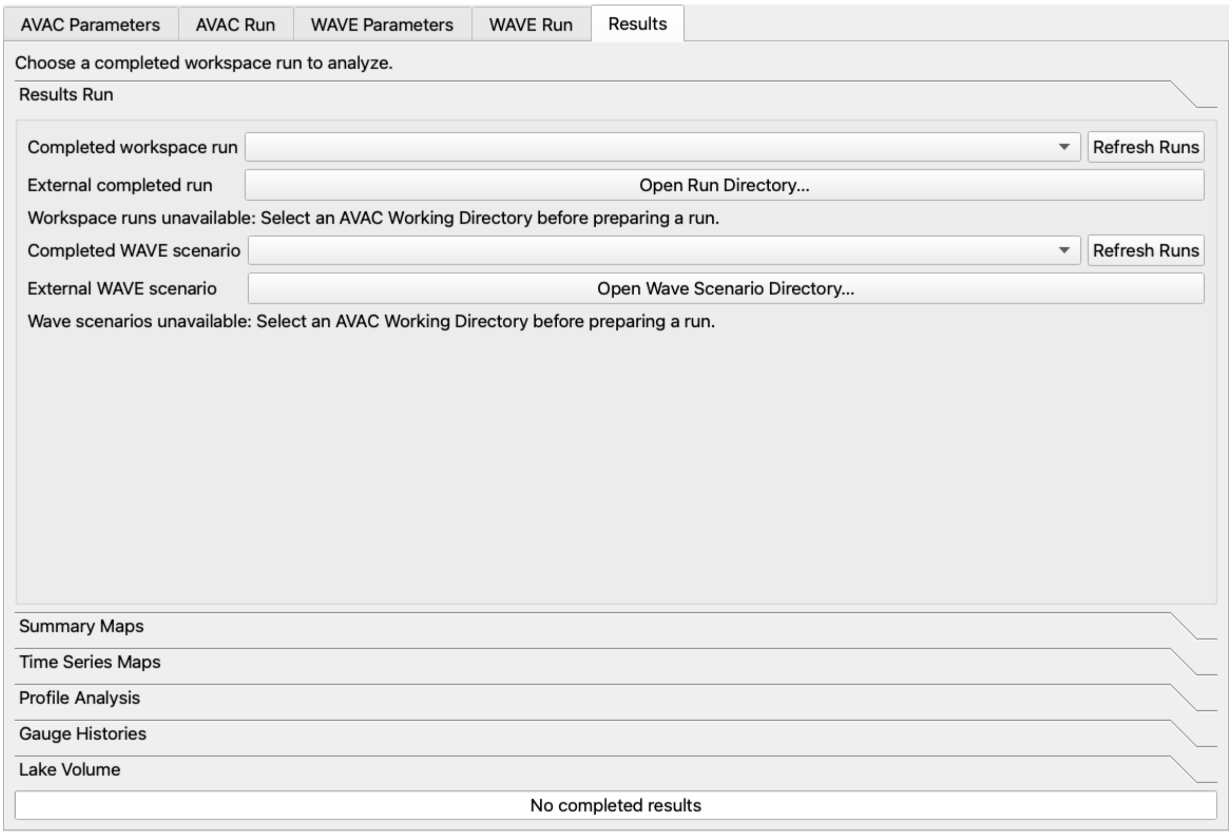

Figure 11: Results Run with the Lake-Wave Extension enabled.

Temporal and derived rasters are stored under the selected AVAC or WAVE run in the Working Directory. Frame-export destinations are chosen by the user; the plugin does not keep generated result rasters in the QGIS application folder.

3.6.4 Profile Analysis

- **Profile layer** lists valid QGIS line layers. **Refresh Layers** refreshes that list.
- **Selected feature** uses the sole usable line automatically. If a layer has multiple lines, select exactly one feature in QGIS; the control reports that feature.
- **Result type** has Selected frame, Full time series (export), and Historical maximum through selected frame. Historical maximum is calculated independently at every sampled point from frame 1 through the current frame; it is not the value of a separate static profile.
- **Variable** selects any available AVAC, lake-clipped AVAC, or WAVE temporal quantity.
- **Sampling** uses raster-cell spacing or enables a user-defined **Sample spacing** in meters.
- **Extract / Plot Profile** samples and plots the chosen result. AVAC depth and snow-surface quantities use the same ground/surface cross-section presentation as WAVE water profiles. **Export CSV** writes the current extraction.
- For Full time series, **Time-series export** writes either one long-form CSV or one PNG profile per frame. PNG profile export is cancellable.

If the selected temporal layer has not been loaded, the profile action loads it first and then resumes automatically.

3.6.5 Gauge Histories

- **Point layer** accepts any valid QGIS point layer and is empty by default. Gauges are postprocessing samples and do not have to be defined before AVAC or WAVE preparation.
- **Variable** independently selects an AVAC or WAVE temporal quantity.
- **Read Gauges from Selected Point Layer** loads a missing result series automatically, transforms point coordinates to the result CRS, and bilinearly samples every usable point in every band.
- **Sampled point** lists the sampled features. **Plot Gauge History** opens the elapsed-time plot and **Export Gauge CSV** writes simulation time and the selected value.

3.6.6 Lake Volume

This accordion appears only when WAVE is enabled. Lake-water volume is calculated at every WAVE output time as water depth times cell area, summed over the initial water-body mask. **Plot Lake Volume History** displays the result and **Export Lake Volume CSV** writes elapsed simulation time and volume in m^3. The interface does not currently expose an outflow or spillway hydrograph.

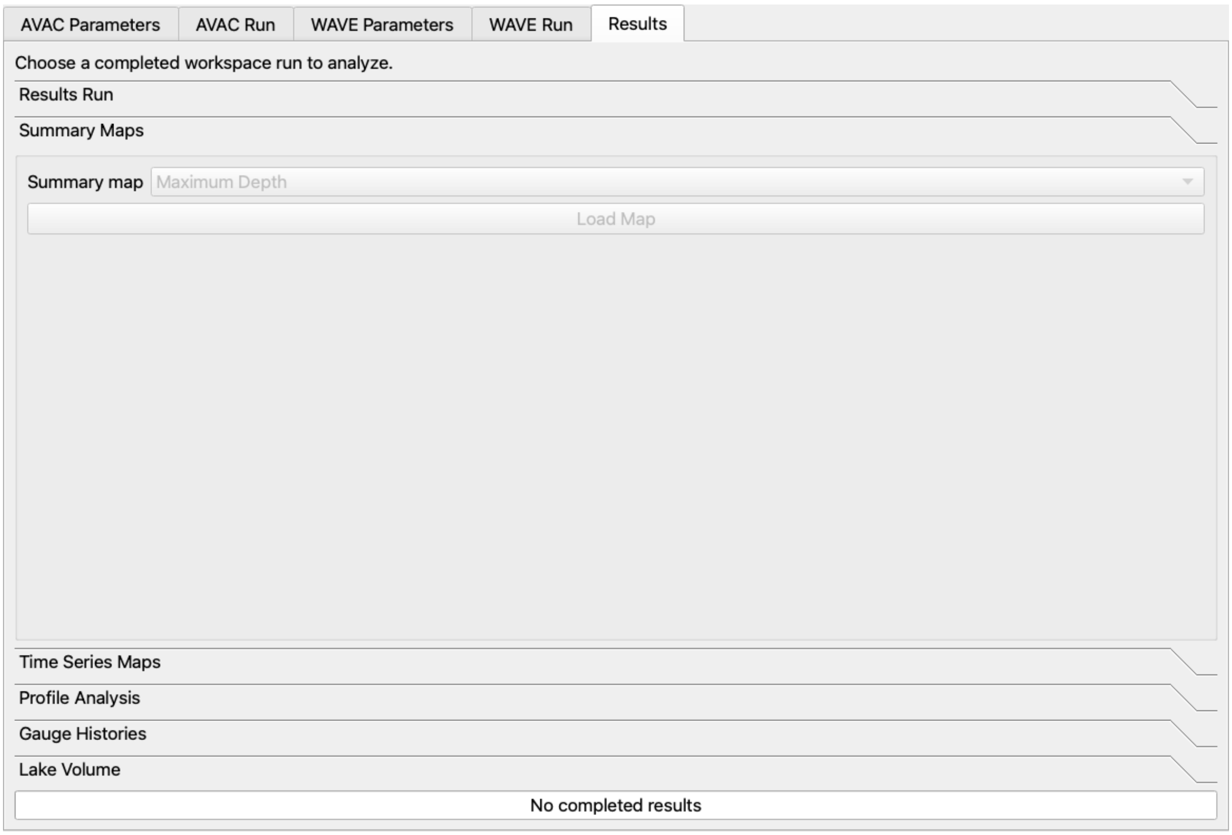

Figure 12: The shared Summary Maps accordion. WAVE choices appear in the selector when the extension is enabled.

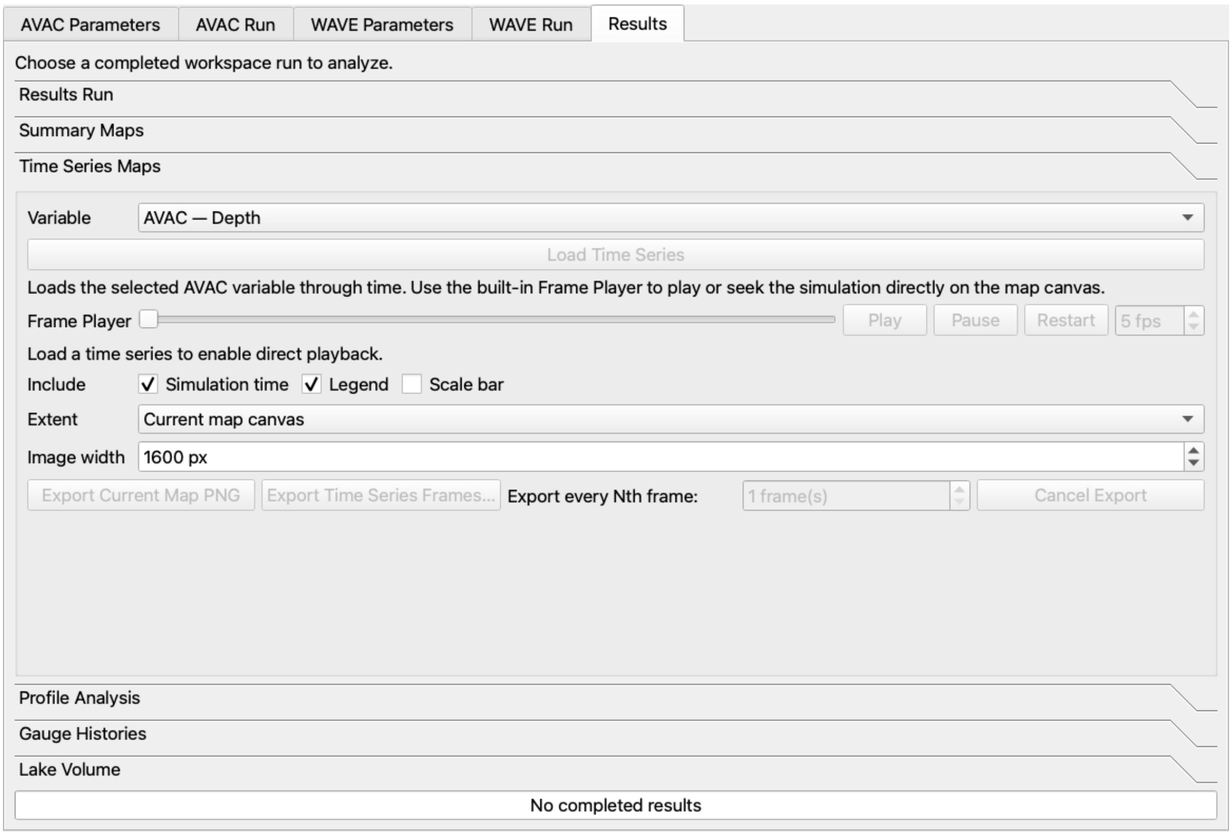

Figure 13: Time Series Maps with the shared Frame Player and PNG export controls.

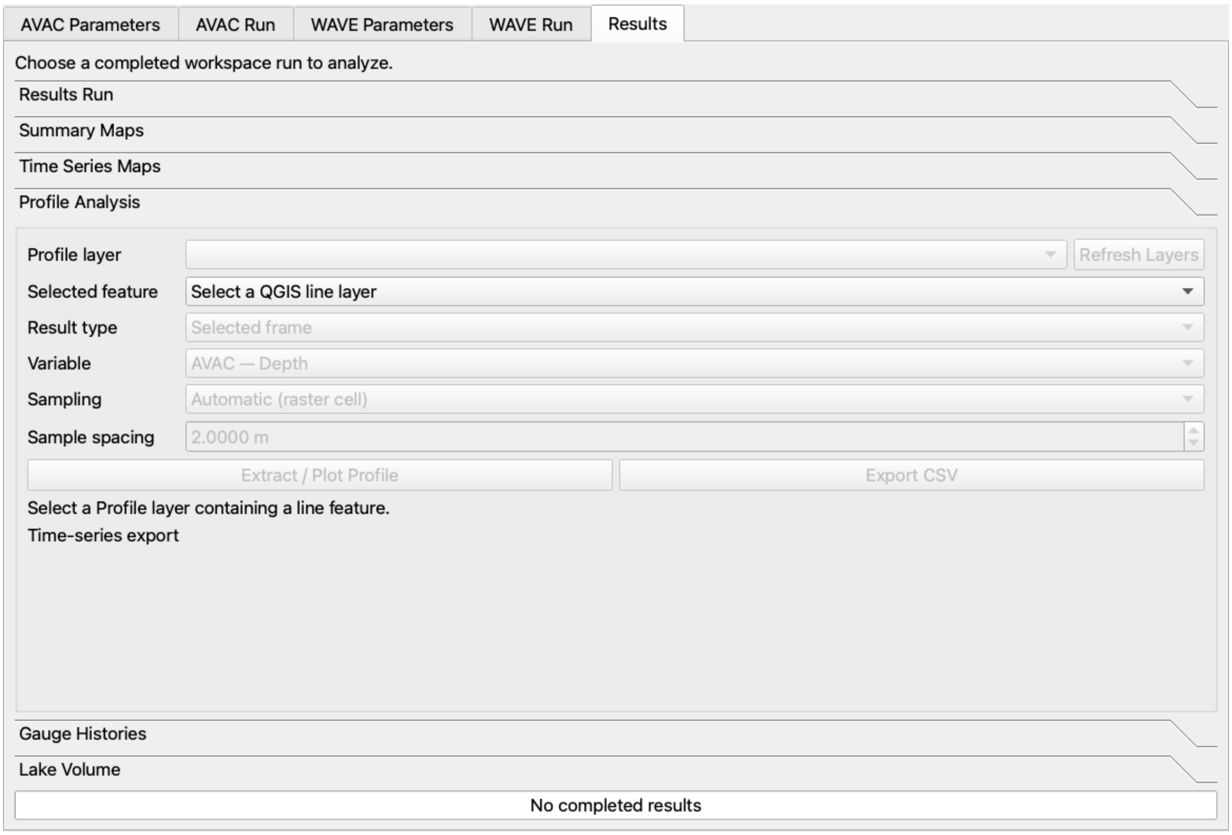

Figure 14: Profile Analysis. The selected-feature row reflects the actual QGIS line selection.

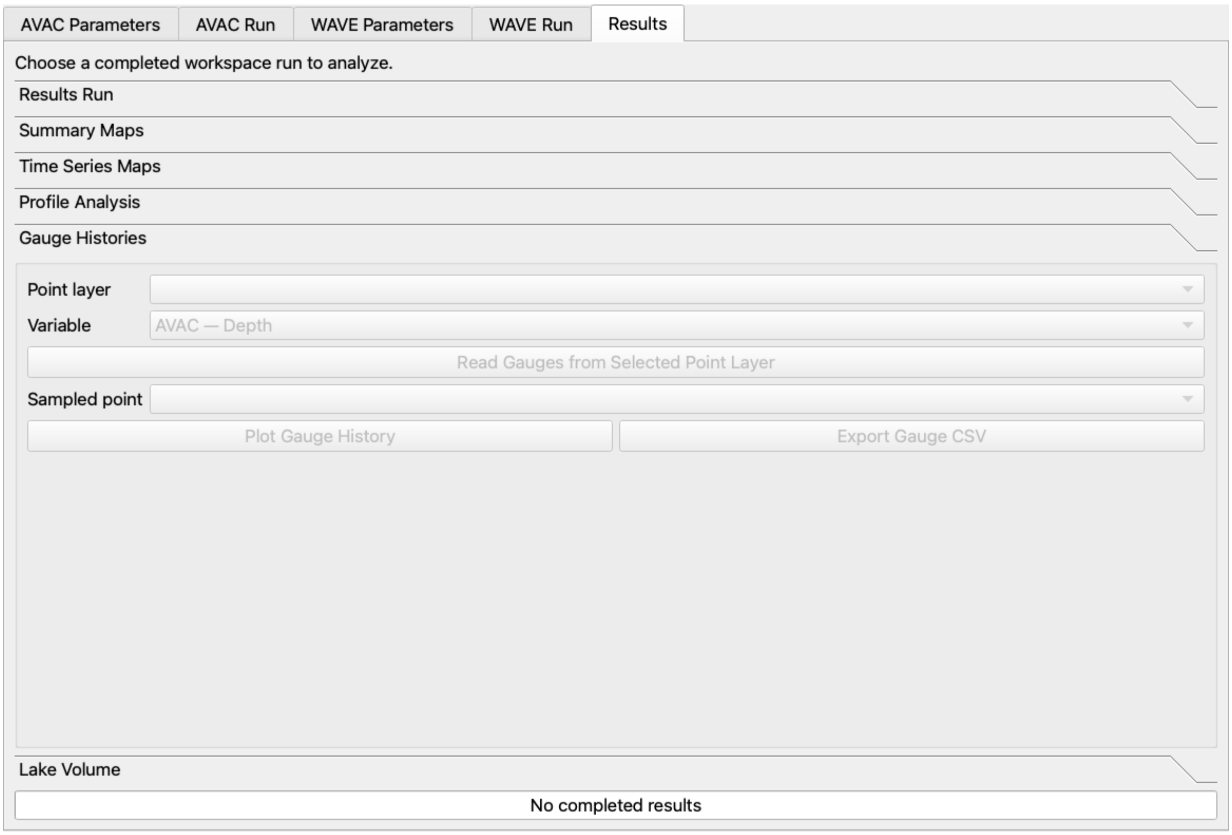

Figure 15: Gauge Histories, shared by AVAC and WAVE and based on post-run QGIS point layers.

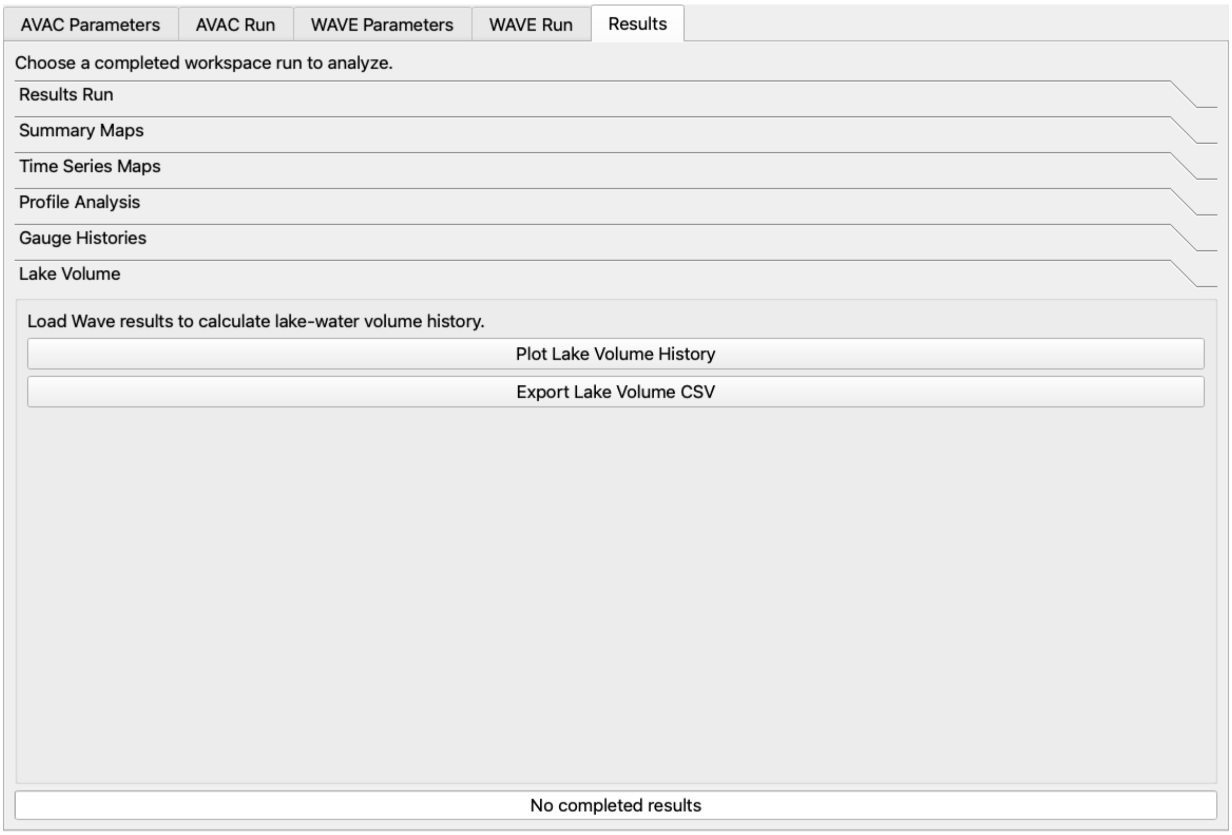

Figure 16: The WAVE-only Lake Volume page in the shared Results tab.

4 Working-directory structure

The Working Directory contains user inputs, isolated runs, derived rasters, and exports. Managed runtimes are installed separately under the user's application-support directory.

```
AVAC Working Directory/
  derived_inputs/                 point-derived lake polygons
  qgis_previews/                  reusable setup preview rasters
  runs/
    run_YYYYMMDD_HHMMSS/
      .avac_qgis_run.json
      inputs/                     materialized source records and files
      Topo/                       AVAC terrain
      AVAC/
        AVAC_configuration.yaml
        init.avacbin                  packaged binary initial depth
        _output/                  raw AVAC solver output
      qgis_results/               AVAC GeoTIFFs and results.json
  wave_runs/
    wave_YYYYMMDD_HHMMSS/
      .avac_qgis_wave_run.json
      impulse_configuration.yaml
      Topo/                       WAVE terrain and force-dry mask
      CL/                         shoreline faces, source, and source ledge
      Wave/_output/               raw WAVE solver output
      qgis_wave_results/          WAVE GeoTIFFs, volume CSV, manifests
```

Run directory names and displayed creation times use the computer's local time. The AVAC and WAVE rasters store elapsed simulation seconds together with one shared per-simulation temporal origin. A WAVE scenario records a reference to its completed AVAC source; it does not overwrite, delete, or silently edit that source run.

5 Portable installation and documentation source

The GitHub source archive does not contain compiled managed runtimes and is not the QGIS installer. Install the platform-specific ZIP from the GitHub Releases page through Plugins > Manage and Install Plugins > Install from ZIP.

This guide is maintained as `docs/ui_reference/AVAC_QGIS_UI_REFERENCE.tex`. Current images are under `docs/ui_reference/images/`; they can be regenerated from the installed QGIS Python environment with `docs/ui_reference/generate_ui_screenshots.py`. Compile from the `docs/ui_reference/` directory with `latexmk -pdf AVAC_QGIS_UI_REFERENCE.t`